

Motorola Xoom

If rumours are to be believed Xoom will hit the markets on February 17 for \$699

Duke Nukem

Finally ending the long wait Duke Nukem will release on May 3

Boot Windows from a virtual drive

Learn how to install and then boot Windows 7 Ultimate from a virtual hard drive located anywhere.

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One of the lesser known facts about Windows 7 is its ability to natively use VHD (Virtual Hard Disk) files and its boot loader can even boot from them. If you have ever used software to mount ISO files as virtual CDs, you will be familiar with the concept of disc images. Unlike .iso files, that store images of a CD / DVD / BluRay discs, VHD files are a way of storing an entire hard drive's structure (partition and data information) inside a file, so they can be used in a virtual machine.

Those who have used Microsoft Virtual PC will be familiar with VHD files,

since it is Microsoft's format of choice for use as virtual hard disks. If you have Windows 7 you can mount VHD files directly, in which case the hard drive will appear with all its partitions just like any external hard drive. What is more exciting and important is that you can have the Windows boot loader boot from an operating system installed in a VHD file. Of course the operating system will need to support this as well; which brings us to an even more exciting fact: Windows 7 Ultimate supports booting from inside a VHD!

What this means is that you can install Windows 7 on a VHD file and boot from it directly, just as Ubuntu Linux does to run from the same partition as you Windows install. You can then have an entire OS contained in a file that you can easily copy, move, backup etc. You can even use a differencing disk, i.e. a disk in which your core OS is stored on one file, and any further changes are stored in another file. You might compare this to running Windows in a virtual machine, however this has significant advantages over that as we will explain at the end of the article.

There are two ways you can do this. The first method is to install Windows 7 onto a VHD file instead of a normal hard disk, and in the second method you can create a VHD file with Windows 7 pre-installed directly from the setup disc without running the setup itself.

For both these steps, we assume that you already have Windows installed on your computer, and are using this method for an additional install.

Option 1: Install Windows to a VHD file

For this you should boot from the Windows 7 Ultimate setup disc,

however instead of starting the setup, start the repair instead.

Click next on "System Recovery Options".

In the list of recovery tools, click on Command Prompt.

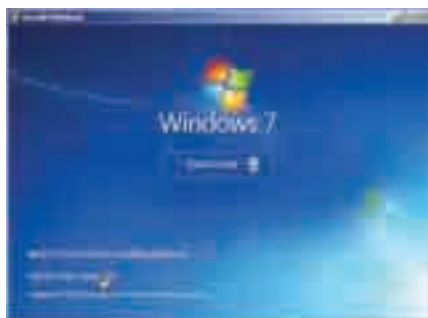
In the command prompt type "diskpart" and hit return to launch the disk partitioning tool. You will need to run the following commands to create a virtual drive in which to install Windows 7:

```
create vdisk file=C:\win-  
dows7.vhd maximum=20000
```

This command will create a new virtual drive at the location specified after "file=" with a maximum size in



Select Command Prompt



Boot from the Windows Ultimate setup disc



Proceed beyond Next

MB specified after "maximum="

```
select vdisk file=C:\win-  
dows7.vhd
```

This command will select the virtual disk file so you can modify it.

```
attach vdisk
```

This will attach the VHD file to it shows up as a normal disk drive.

```
exit
```

This will exit the diskpart program.

Now you can manually start the



Attach command is used to attach the VHD file